

```

public class BinarySearchIterative{
    public static int binarySearch(int[]arr,int target){
        int left = 0;
        int right = arr.length - 1;
        while(left <= right) {
            int mid = left + (right - left) / 2;
            if(arr[mid] == target){
                return mid;
            }
            if(arr[mid] < target){
                left = mid + 1;
            }
            else{
                right = mid - 1;
            }
        }
        return -1;
    }
    public static void main(String[]args){
        int[] sortedArray = {2,5,8,12,16,23,38,56,72,91};
        int target = 23;
        int result = binarySearch(sortedArray,target);
        if(result == -1){
            System.out.println("Element not found.");
        }else{
            System.out.println("Element found at index:" + result);
        }
    }
}

```